ZCAS UNIVERSITY

SCHOOL OF COMPUTING, TECHNOLOGY & APPLIED SCIENCES

CIT3382 CRITICAL INFRASTRUCTURE PROTECTION

FINAL EXAMINATION

DATE: 25TH NOVEMBER 2024

TIME: 14:00 – 17:00

TIME ALLOWED: WRITING – THREE HOURS

READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: This question is **compulsory** and must be attempted.
2. Section B: Answer **THREE (3)** questions from this section.
3. This examination paper carries a total of **100 marks.**
4. Candidates must **not turn this page** until the invigilator tells them to do so.

Section A: All questions in this section are compulsory and must be attempted.

QUESTION 1

SCENARIO

A terrorist group threatens to disrupt operations at a major financial institution by launching a coordinated cyber and physical attack. The institution is a key player in the nation's financial ecosystem.

1) Identify the types of critical infrastructure systems that would be impacted by such an attack. **[5 Marks]**
2) Recommend a multi-layered security approach to protect the institution from both physical and cyber threats. **[5 Marks]**
3) Discuss how government agencies and private entities should collaborate to address this threat. **[5 Marks]**
4) How would a successful attack on the institution affect the broader national financial system and economy? **[5 Marks]**
5) What security measures would you put in place to defend against both cyber and physical threats, and are they sufficient? **[5 Marks]**
6) What contingency and recovery plans would you put in place to mitigate damage and restore operations in the event of an attack? **[5 Marks]**
7) How should the institution handle communication with the public and media to prevent panic while addressing the threat? **[5 Marks]**
8) What steps can be taken to enhance resilience against similar threats in the future? **[5 Marks]**

TOTAL MARKS 40

Section B: Answer any **THREE** questions only.

QUESTION 2

Define the following terms and provide one example for each:

a. Critical Infrastructure

b. Vulnerability

c. Risk Management

d. Cyber Terrorism

[20 Marks]

QUESTIONS 3

The energy sector has been a frequent target of cyber-attacks. Using examples

a) Describe the consequences of a cyber-attack on the energy sector. **[5 Marks]**

b) Analyse the 2015 Ukrainian power grid cyber-attack and highlight the vulnerabilities exploited. **[5 Marks]**

c) State 5 critical infrastructure sectors that are at risk of terror attacks except the energy sector. **[10 Marks]**

[20 Marks]

QUESTION 4

a) List and describe the 5 types of terrorism. **[10 Marks]**

b) List and describe 5 of the terrorism tactics that may occur. **[10 Marks]**

QUESTION 5

a) Based on the "Hatman" attack on a Saudi petrochemical facility, discuss the role of safety systems in mitigating the impact of cyber threats. **[10 Marks]**

b) Explain the importance of categorizing critical infrastructure into "soft" and "hard" targets and give two examples of each. **[10 Marks]**

END OF EXAM